FF-N-180a
February 3, 1966
SUPERSEDING
Fed. Spec. FF-N-180
November 2, 1955

FEDERAL SPECIFICATION

NEEDLES, EXCEPT SURGICAL, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand needles, except surgical.

1.2 Classification.

1.2.1 *Types, classes, and styles.* Hand needles covered by this specification shall be of the following types, classes, and styles as specified (see 6.2)

Type I—Straight.
  Class 1—Round point.
    Style B—Upholsterer's (double point).
    Style D—Harness.
  Class 2—Three-square point, glover's.
  Class 3—Diamond point, packing.

Type II—Bent (at point only), packing.

Type III—Curved.
  Class 1—Round point (regular).
  Class 2—Three-square point.

1.2.2 *Size and duty.* The size and duty (extra light, light, or heavy) of the hand needles shall be as listed in tables of section 3 under the respective types, classes, and styles, and as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging Of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample of the finished specified needles covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production needles.

3.2 Materials. The material used in the manufacture of hand needles covered by this specification shall be of tool steel normally used in commercial practice. The needles shall be hardened and tempered in accordance with the best commercial practice for this type of commodity.

3.3 Construction. Each hand needle shall be ground to a sharp fine point blending in shape with the body of the needle directly in back of the point. The spreading edges of the triangular and diamond sections (when applicable) shall be ground and rounded in a manner to prevent material fibers from being cut when pierced by the needle in the normal sewing operation. The eye of the needles shall be centered between the lateral edges and shall be designed to prevent cutting of the thread or cord in ordinary use. Types I and II needles shall not break or take a permanent set when subjected to a horizontally applied force as specified in 4.4.1.1 and 4.4.1.2, as applicable. Type III needles shall not break or take a permanent set when a force is applied simultaneously to the eye and point as specified in 4.4.1.3.

3.3.1 *Type I, straight.*

3.3.1.1 *Class 1, round point.*

3.3.1.1.1 *Style B, double, upholsterer's* Style B needles shall conform to table I and shall be similar to figure 1.

TABLE I. *Type I, class 1, style B, straight, round point, double, upholsterer's*

Size (length L) ± 1/8 inch	Section BB diameter (± 0.005 inch)		
	Extra light	Light	Heavy
Inches	*Inch*	*Inch*	*Inch*
8	—	0.080	0.092
10	0.080	.092	.104
12	.080	.092	.104
14	.080	.092	.104
16	.080	.104	.116

3.3.1.1.2 *Style D, harness.* Style D needles shall be furnished in the commercial sizes shown in table II and shall be similar to figure 2.

TABLE II. *Type I, class 1, style D, straight, round point, harness*

Commercial size	Length L	Section BB diameter (± 0.005 inch)
	Inches	*Inch*
0	2- 1/4 —2- 1/2	0.048
2	2- 1/16—2- 3/16	.040
3	2 —2- 1/8	.040
4	1-13/16—1-15/16	.036
6	1- 9/16—1- 3/4	.032

3.3.1.2 *Class 2, three-square point, glover's.* Glover's needles shall be furnished in the commercial sizes shown in table III and shall be similar to figure 3.

TABLE III. *Type I, class 2, straight, three-square point, glover's*

Commercial size	Length L	Section BB diameter (± 0.005 inch)
	Inches	*Inch*
000	2- 1/16—2- 1/8	0.064
00	2- 1/16—2- 1/8	.051
0	1- 7/8 —2	.051
1	1-13/16—1-15/16	.048
2	1-13/16—1 -7/8	.043
3	1- 9/16—1-11/16	.040
4	1- 5/8 —1-11/16	.036
6	1- 1/2 —1- 5/8	.028
8	1- 1/4 —1- 3/8	.024

3.3.1.3 *Class 3, diamond point, packing.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

Class 3 needles shall conform to table IV and shall be similar to figure 4.

TABLE IV. *Type I, class 3, straight, diamond point, packing*

Size (length L) ± 1/8 inch	Section BB diameter (± 0.005 inch)
Inches	*Inch*
6	0.144

3.3.2 *Type II, bent, packing.* Type II needles shall conform to table V and shall be similar to figure 5.

TABLE V. *Type II, bent, packing*

Size (length L) ± 1/8 inch	Section BB diameter (± 0.005 inch)
Inches	*Inch*
3	0.080
4	.104
5	.128
6	.144
7	.160

3.3.3 *Type III, curved.*

3.3.3.1 *Class 1, round point (regular).* Class 1 needles shall conform to table VI and shall be similar to figure 6.

TABLE VI. *Type III, class 1, curved round point, regular*

Size (length L) ± 1/8 inch	W (approximate)	Section BB diameter (± 0.005 inch)		
		Extra light	Light	Heavy
Inches	*Inches*	*Inch*	*Inch*	*Inch*
2-1/2	1-3/4	0.036	0.048	—
4	2-3/4	.048	.064	0.072
4-1/2	3	—	.064	.072
5	3-3/8	—	.064	.080
8	5-1/4	—	.072	.092

3.3.3.2 *Class 2, three-square point.* Class 2 needles shall conform to table VII and shall be similar to figure 7.

TABLE VII. *Type III, class 2, curved, three-square point*

Size (length L) ± 1/8 inch	W (approximate)	Section BB diameter (± 0.005 inch)		
		Extra light	Light	Heavy
Inches	*Inches*	*Inch*	*Inch*	*Inch*
3	2	0.040	0.048	0.064
5	3-3/8	—	.064	.080

3.4 **Finish.** The needle surfaces shall be nickel plated in accordance with standard commercial practice for this type of commodity.

3.5 **Workmanship.** The finished needles shall be smooth and free of burrs, slivers, nicks, pitholes, and scratches.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Preproduction sample inspection.** When a preproduction sample is required the sample shall be examined for defects in table VIII and specified dimensions. The preproduction sample shall be tested as specified in 4.4.1.

4.3 **Sampling for inspection and acceptance.** Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 *Component and material inspection.* In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 *Material examination.* An examination shall be made to determine that steel and heat treatment requirements of 3.2, and nickel plating requirements of 3.4 are complied with.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

4.3.2 *End item inspection.* The lot shall consist of all the needles of one type, class, style, and size offered for inspection at one time. The sample unit shall be one needle.

4.3.2.1 *Visual examination.* Examination shall be made of the end item for defects in table VIII. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE VIII. *Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Finish	Surface (including periphery of the eye) is not smooth	X	
	Evidence of imbedded foreign matter		X
Construction and workmanship	Does not have fine sharp point	X	
	Edge of triangular or diamond section (when applicable) not ground and rounded		X
	Eye not centered between lateral edges	X	
	Eye is closed	X	
	Eye is partially closed		X
	Sharp edge at eye (inside or outside)	X	
	Burrs or slivers not removed	X	
	Needle is cracked, broken, or fractured	X	
	Evidence of pitholes, nicks, or scratches		X

4.3.2.2 *Dimensional inspection.* Inspection shall be made of needles for compliance with dimensions specified. Any deviation therefrom shall constitute a defect. The inspection level shall be S-2 with an AQL of 6.5 expressed in terms of defects per hundred units.

4.3.2.3 *Testing of end item.* Needles shall be tested a minimum of 3 times as specified in 4.4.1. Any deviation therefrom shall constitute a defect. The inspection level shall be S-4 with an AQL of 2.5 expressed in terms of defects per hundred units.

4.3.3 *Examination of preparation for delivery.* Examination shall be made for preservation, packaging, packing, and marking in accordance with MIL-H-15424.

4.4 Tests.

4.4.1 *Deflection test.*

4.4.1.1 *Types I and II needles up to and including 7 inches in length.* The point of the needle shall be held rigidly by a vise or other means and shall be gripped not more than 7/32 inch from the point as shown on figures 8 and 9. A force shall be applied to the eye end of the needle as close as possible to the heel, in a direction that will cause the needle to bend while the point remains stationary. A relatively sharp-edge tool such as a screwdriver or chisel shall be used to apply the force. Needles up to and including 7 inches in length shall be deflected a linear distance in accordance with table IX and figure 8. This test shall be performed for each of the three planes represented by the triangular section of type I, class 2, three-square point needles and for each of the four planes represented by the diamond section of type I, class 3 and type II needles.

4.4.1.2 *Type I needles from 8 inches up to and including 16 inches in length.* The needles shall be tested as specified in 4.4.1 except that the force shall be applied as shown on figure 9 and the deflection of the needles shall be in accordance with table X.

4.4.1.3 *Type III needles.* A force shall be applied simultaneously to the point and eye of the needle causing a reduction in length "W" (see fig. 6) of at least 25 percent to determine compliance with 3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

TABLE IX. *Deflections for needles up to and including 7 inches in length*

Size (length L) inches	Deflection for sectional sizes BB (inches)													
	0.024	0.028	0.032	0.036	0.040	0.043	0.048	0.051	0.064	0.080	0.104	0.128	0.144	0.160
1-5/16	1/8	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....
1-9/16	.....	5/32	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....
1-5/8	.....	.....	.....	.....	1/8	.....	.....	.....	.....	.....	.....	.....	.....	.....
1-11/16	.....	.....	3/16	5/32	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....
1-7/8	.....	.....	.....	7/32	.....	3/16	5/32	.....	.....	.....	.....	.....	.....	.....
2	.....	.....	.....	.....	.....	.....	.....	5/32	.....	.....	.....	.....	.....	.....
2-1/16	.....	.....	.....	.....	1/4	.....	.....	.....	.....	.....	.....	.....	.....	.....
2-1/8	.....	.....	.....	.....	1/4	.....	.....	3/16	5/32	.....	.....	.....	.....	.....
2-3/8	.....	.....	.....	.....	.....	.....	9/32	.....	.....	.....	.....	.....	.....	.....
2-1/2	.....	.....	.....	.....	.....	.....	5/16	.....	.....	.....	.....	.....	.....	.....
3	.....	.....	.....	.....	.....	.....	.....	.....	.....	9/32	.....	.....	.....	.....
4	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	3/8	.....	.....	.....
5	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	1/2	.....	.....
6	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	21/32	.....
7	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	.....	13/16

TABLE X. *Deflections for needles of 8 inches up to and including 16 inches in length*

Size (length L) inches	Deflection for sectional sizes BB (inches)				
	0.072	0.080	0.092	0.104	0.116
8	1-7/32	1-1/16	15/16	.....	.....
10	.....	1-15/16	1-9/32	1-1/8	.....
12	.....	1-29/32	1-21/32	1-7/16	.....
14	.....	2-5/16	2	1-25/32	.....
16	.....	2-23/32	.....	2-3/16	1-7/8

5. PREPARATION FOR DELIVERY

For civil agency procurements, the definitions and application of levels of preservation, packaging, and packing shall be in accordance with Fed. Std. No. 102.

5.1 Levels A, B, and C. Needles shall be preserved, packaged, and packed for each level of protection, in accordance with the applicable requirements of MIL-H-15424 (see 6.2).

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The needles covered by this specification are intended for general use in sewing, and for packaging and packing various items.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

(*b*) Type, class, and style needle required (see 1.2.1).
(*c*) Size and duty required (see 1.2.2).
(*d*) When a preproduction sample is required (see 3.1).
(*e*) Selection of applicable levels of packaging and packing (see 5.1).

6.3 For civil agency procurements. When level B packaging is required for civil agency procurements, the level A requirements in 8.3.1 of MIL-H-15424 shall apply.

6.4 Supersession data. The following type, class, style, and size needles specified in FF-N-180 have been deleted in this revision as there are no known Federal users.

(a) Type I, class 1, style A.
(b) Type I, class 1, style B, sizes 6, 18, 20.
(c) Type I, class 1, style C.
(d) Type I, class 2, style A.
(e) Type I, class 2, style B.
(f) Type I, class 2, style C.
(g) Type I, class 2, style D.
(h) Type I, class 3, sizes 4, 5.
(i) Type II, sizes 3-1/2, 4-1/2, 5-1/2, 8.
(j) Type III, class 1, style A.
(k) Type III, class 1, style B, sizes 2, 3, 3-1/2, 6, 7, 9, 10.
(l) Type III, class 2, sizes 2, 2-1/2, 3-1/2, 4, 6, 7, 8, 9, 10.
(m) Type III, class 3.

6.5 Transportation descriptions. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Needles, not otherwise indexed by name.
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Needles, not otherwise indexed.
Truckload minimum weight 20,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—GL
Navy—SA
Air Force—67

Review activities:

Navy—SA
Air Force—67

User activities:

Navy—MC, YD

Preparing activity:

Army—GL

Review and User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing Of DOD Standardization Documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

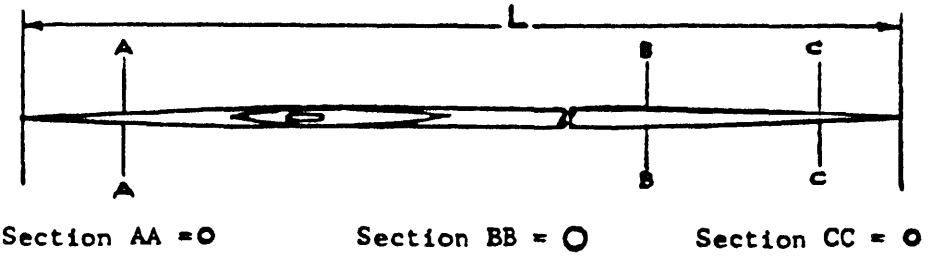


FIGURE 1. Type I, class 1, style B, straight, round point, double, upholsterer's.

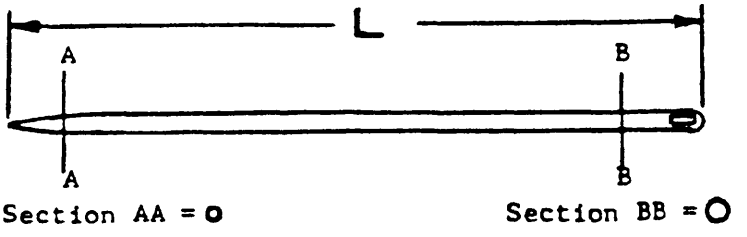


FIGURE 2. Type I, class 1, style D, straight, round point, harness.

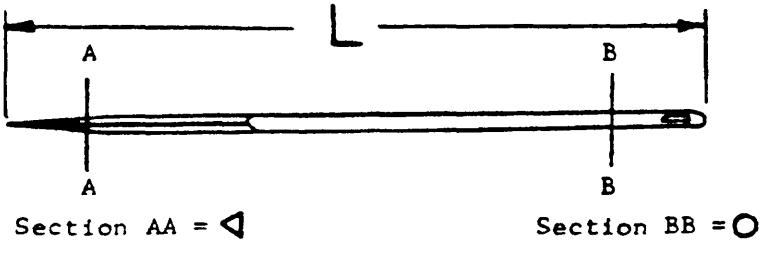


FIGURE 3. Type I, class 2, straight, three-square point, glover's.

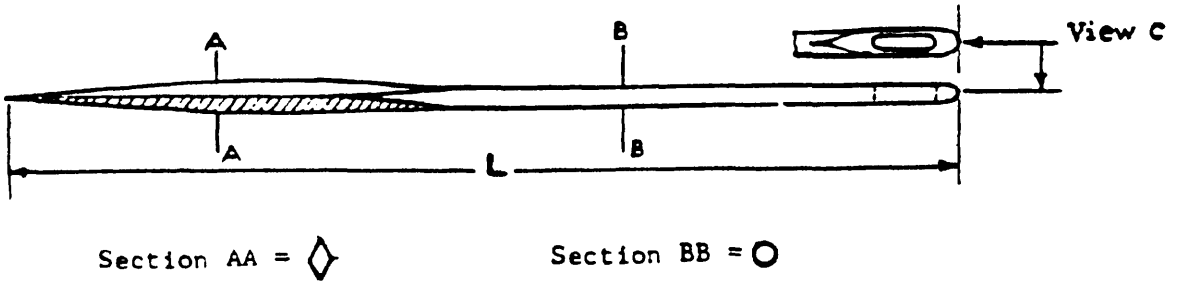


FIGURE 4. Type I, class 3, straight, packing.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:46Z
Check the source to verify that this is the currentversion before use.

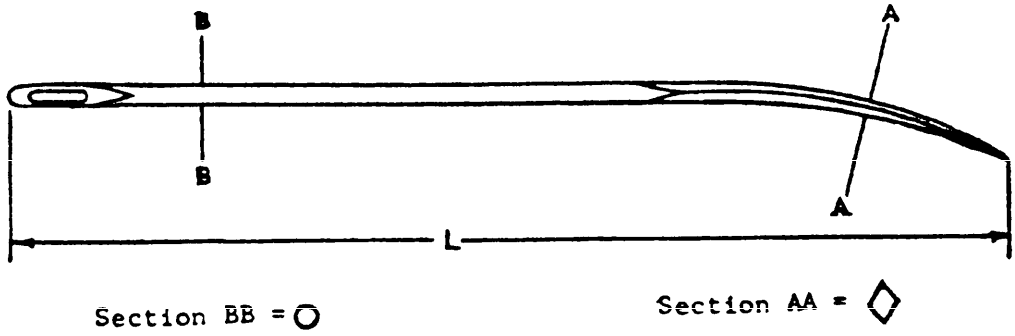


FIGURE 5. Type II, bent, packing.

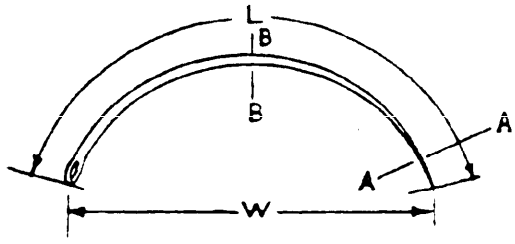


FIGURE 6. Type III, class 1, curved, round point, regular.

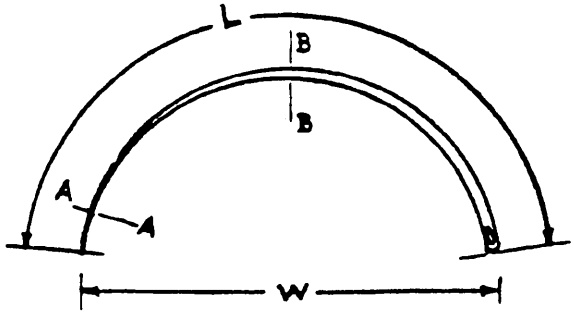


FIGURE 7. Type III, class 2, curved, three-square point.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

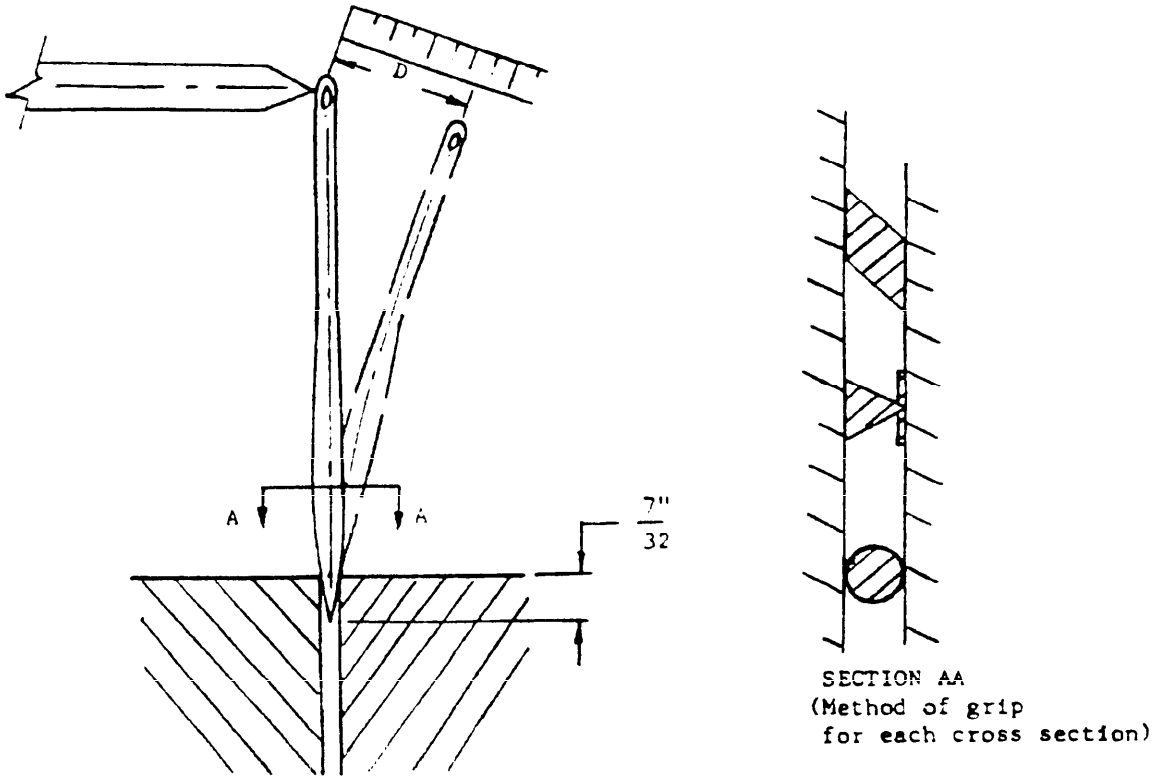


FIGURE 8. Deflection test for needles up to and including 7 inches in length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

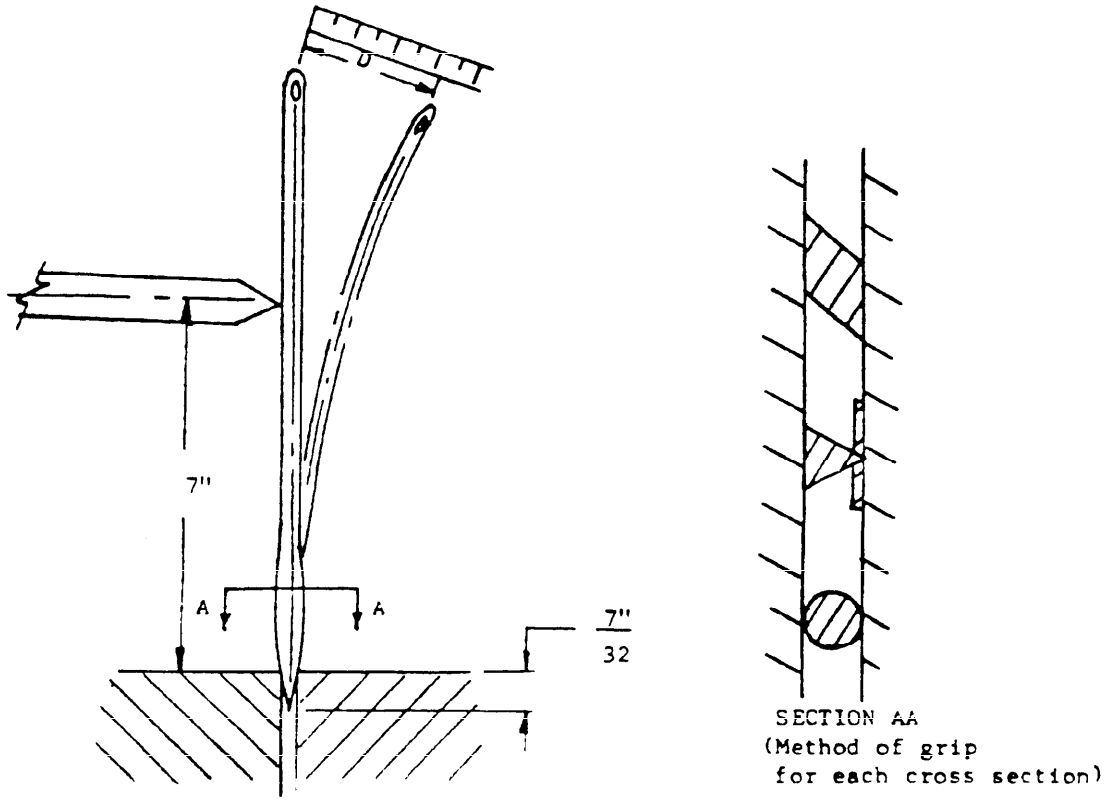


FIGURE 9. Deflection test for needles of 8 up to and including 16 inches in length.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

"to detach this form, cut along this line"

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Needles, Except Surgical, Hand - FF-N-180a

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01762

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRE-SS
Natick, Massachusetts 01762

FOLD

"to detach this form, cut along this line"

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:46Z
Check the source to verify that this is the current version before use.